

# RIYA PATEL

Bharuch, Gujarat, India | +91 9510177953 | [rdp2245@gmail.com](mailto:rdp2245@gmail.com)

[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

## EDUCATION

### B.Tech: Computer Science & Engineering

2023 – 2027

P P Savani University, Surat, Gujarat

Coursework: Data Structures & Algorithms · DBMS · Operating Systems · Computer Networks · OOP

## SKILLS

- **Languages:** Python, Java, JavaScript (ES6+), HTML, CSS, SQL
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Backend:** FastAPI, Spring Boot, REST APIs
- **Databases:** PostgreSQL, Redis, SQL
- **Tools:** Docker, Git, AWS (EC2, S3, RDS, IAM)

## PROJECTS

### Adaptive System — Reliability Monitoring & Adaptive Recovery Platform — [github.com/Riyadpatel24/adaptive\\_system](https://github.com/Riyadpatel24/adaptive_system)

Reliability monitoring platform with real-time telemetry, failure prediction, and fault-injection testing.

- Built a recovery loop: telemetry ingestion → signal normalisation → health classification → LOCKDOWN/THROTTLE actions via safety guard and cooldown manager
- Failure prediction from rolling risk histories and root cause analysis via DependencyGraph; PolicyEngine adapts timeout and retry limits based on live failure rates
- Implemented fault-injection testing using CPU spike and memory leak simulations to validate recovery policies and health classification under failure conditions; FastAPI daemon thread serves live snapshots at /snapshot for real-time observability

### Sudoku Masters — [github.com/Riyadpatel24/sudoku-master](https://github.com/Riyadpatel24/sudoku-master)

Browser-based Sudoku game built with React, TypeScript, Vite, and Tailwind CSS.

- Built a full 9×9 Sudoku board with input validation, notes mode (pencil marks), and on-screen number pad; responsive across desktop and mobile
- Implemented puzzle generation and solving logic in TypeScript (utils/sudoku.ts) with a modular component architecture separating gameplay, UI, and utilities
- Added contest mode (timed competitive gameplay), daily streak tracking, and a leaderboard modal; integrated confetti animation on puzzle completion via canvas-confetti
- Tech stack: React 18, TypeScript, Vite, Tailwind CSS v4, lucide-react

### VoiceNotes Organizer — AI Audio-to-Notes Pipeline — [github.com/Riyadpatel24/voicenotes-organizer](https://github.com/Riyadpatel24/voicenotes-organizer)

Full-stack pipeline: audio upload → Whisper transcription → LLM-structured notes, tasks, and decisions.

- React + FastAPI pipeline: audio transcribed via Groq Whisper Large V3, then structured by LLaMA 3.3 70B into summaries, action items, deadlines, and decisions in JSON
- Modular architecture cleanly separates transcription and structuring layers — AI models swappable without changing any downstream logic

## CERTIFICATIONS

### IBM Java Developer Professional Certificate

Jun 2025

Coursera / IBM · 11 courses · Spring Boot, Microservices, Docker, REST APIs, NoSQL/SQL, Generative AI

### AWS Academy Graduate — Cloud Foundations

Nov 2025

Amazon Web Services · 20 hours · EC2, S3, RDS, IAM, VPC, cloud security fundamentals

## LANGUAGES

**English:** Professional (C1) | **Hindi:** Native | **Gujarati:** Native | **German:** Elementary (A1) | **Mandarin Chinese:** Elementary (A2)